

Continuous Random Variables

Sourav Karmakar

Outline

- 1 Continuous Random Variables
- 2 Expectations and Variance
- 3 Common Continuous Distributions
- 4 Quiz

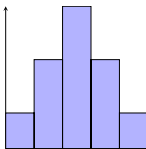
Chapter 1: Continuous Random Variables

Transition: Discrete to Continuous (Histograms to Curves)

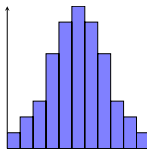
Imagine measuring the height of 10,000 people.

- As we increase the precision of our measurement (bins get narrower), the histogram approaches a smooth curve.
- In the limit (infinite precision), this curve becomes the **Probability Density Function (PDF)**.

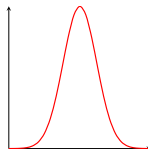
Coarse (Nearest 5cm)



Fine (Nearest 1mm)



Continuous (Infinite Precision)

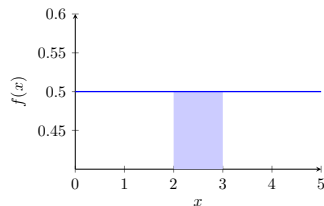


Probability Density Function (PDF)

For Continuous Random Variables, we use a **Probability Density Function (PDF)**, denoted $f(x)$.

Key Differences from PMF

- 1 **PMF (Discrete):** $P(X = x)$ is the height of the bar.
- 2 **PDF (Continuous):** The height $f(x)$ is **NOT** the probability.
- 3 **Area is Probability:** Probability is the **Area under the curve**.



Calculating Probability (Integration)

To find the probability that X falls between a and b , we integrate the PDF:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Properties of PDF

- ① $f(x) \geq 0$ for all x . (Density cannot be negative).
- ② Total Area = 1: $\int_{-\infty}^{\infty} f(x) dx = 1$.

The Zero Probability Paradox

Concept: For a continuous variable, the probability of taking an **exact** value is Zero.

$$P(X = c) = \int_c^c f(x) dx = 0$$

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Intuition: The Infinite Digital Scale Imagine a digital scale with infinite decimal places.

- You weigh an apple.
- Is it **exactly** 100g?
- It might be 100.000000001g or 99.999999998g.
- The probability of hitting exactly 100.000... (infinite zeros) is impossible (1 outcome out of infinite possibilities).

Conclusion: We only calculate probabilities for Intervals (e.g., between 99.0g and 101.0g).

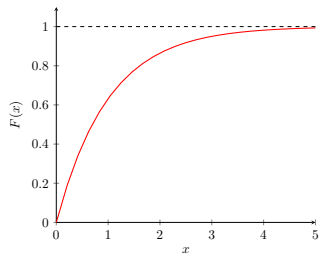
Cumulative Distribution Function (CDF)

The CDF, $F(x)$, gives the probability that the variable takes a value less than or equal to x .

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

Relationship:

- PDF is the derivative of CDF: $f(x) = F'(x)$.
- Prob of interval:
 $P(a \leq X \leq b) = F(b) - F(a)$.



Numerical Example 1: The Linear PDF

Problem: Let X be a random variable with PDF:

$$f(x) = \begin{cases} 2x & \text{for } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Calculate $P(X > 0.5)$.

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Solution:

$$\begin{aligned} P(X > 0.5) &= \int_{0.5}^1 2x \, dx \\ &= [x^2]_{0.5}^1 \\ &= (1^2) - (0.5^2) = 1 - 0.25 = 0.75 \end{aligned}$$

Numerical Example 2: The Constant PDF

Problem: A bus arrives randomly between 8:00 and 8:10. Let X be the minutes past 8:00. PDF: $f(x) = 0.1$ for $0 \leq x \leq 10$. Find the probability the bus arrives in the first 3 minutes.

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Solution:

$$P(0 \leq X \leq 3) = \int_0^3 0.1 \, dx$$

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Solution:

$$P(0 \leq X \leq 3) = \int_0^3 0.1 \, dx$$

$$= [0.1x]_0^3$$

$$= 0.1(3) - 0.1(0) = 0.3$$

(30% chance)

Chapter 2: Expectations and Variance

Expectation (Mean) in Continuous Domain

For discrete, we summed $x \cdot P(x)$. For continuous, we integrate:

Formula

$$E[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

It represents the “Center of Gravity” of the probability density.

Numerical Example: Expectation

Problem: Find the Mean of $f(x) = 2x$ on $[0, 1]$.

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Solution:

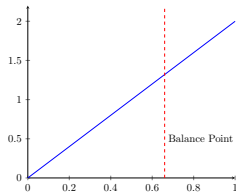
$$\begin{aligned} E[X] &= \int_0^1 x \cdot (2x) dx = \int_0^1 2x^2 dx \\ &= \left[\frac{2x^3}{3} \right]_0^1 \end{aligned}$$

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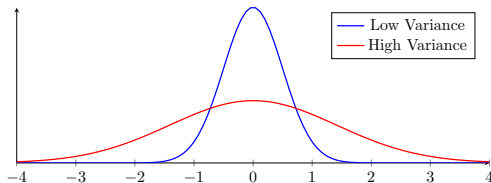
Variance and Standard Deviation of Continuous RV

Variance ($Var(X)$): The average squared distance from the mean. Let $\mu = E[X]$

$$Var(X) = E[(X - \mu)^2] = \int (x - \mu)^2 f(x) dx \quad \text{OR} \quad E[X^2] - (E[X])^2$$

Standard Deviation (σ): A measure of spread in original units.

$$\sigma = \sqrt{Var(X)}$$



Numerical Example: Variance

Problem: Find variance for $f(x) = 2x$ on $[0, 1]$ (We know $\mu = 2/3$).

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Step 1: Calculate $E[X^2]$

$$E[X^2] = \int_0^1 x^2 \cdot (2x) dx = \int_0^1 2x^3 dx$$

$$= \left[\frac{2x^4}{4} \right]_0^1 = \frac{1}{2} = 0.5$$

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Step 2: Apply Formula

$$\text{Var}(X) = 0.5 - (2/3)^2 = 0.5 - 0.444 = 0.055$$

Linearity Properties

Expectation: $E[aX + b] = aE[X] + b$. **Variance:** $Var(aX + b) = a^2Var(X)$.

Real World Example: Project Cost Estimation

- Let X be the time (in hours) to complete a coding task.
- Suppose $E[X] = 10$ hours and standard deviation $\sigma_X = 2$ hours ($Var(X) = 4$).
- You hire a consultant who charges **\$50 per hour** plus a **\$100 fixed fee**.
- Cost function: $Y = 50X + 100$.

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- Cost function: $Y = 50X + 100$.

Expected Cost: $50(10) + 100 = \$600$.

Variance of Cost: $50^2 \cdot Var(X) = 2500 \cdot 4 = 10,000$.

Std Dev of Cost: $\sqrt{10,000} = \$100$.

(Notice the fixed fee \$100 does not affect the uncertainty/spread).

Chapter 3: Common Distributions

1. Uniform Distribution

Notation: $X \sim U(a, b)$

Description: All intervals of the same length are equally probable.

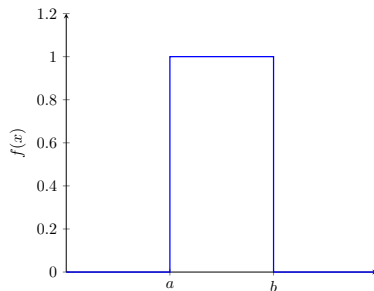
Parameters:

- a (The Minimum): The lower bound of the distribution.
- b (The Maximum): The upper bound of the distribution.

- **PDF:** $f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$.

- **Mean:** $\frac{a+b}{2}$

- **Variance:** $\frac{(b-a)^2}{12}$



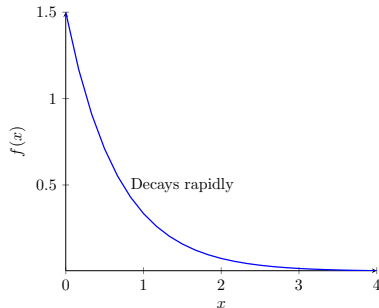
2. Exponential Distribution

Notation: $X \sim \text{Exp}(\lambda)$

Use Case: Modeling time until an event (e.g., time until next customer arrives).

Parameter: λ : commonly known as the rate parameter.

- **PDF:** $f(x) = \lambda e^{-\lambda x}$ ($x \geq 0$)
- **Mean:** $1/\lambda$
- **Variance:** $1/\lambda^2$



Exponential: Memoryless Property

The Property

$$P(X > s + t \mid X > s) = P(X > t)$$

Interpretation: The probability of waiting t more minutes is the same, regardless of how long (s) you have already waited.

Real Life Example:

- **Radioactive Decay:** An atom doesn't "age". The chance it decays in the next second is constant, whether it's 1 second old or 1 billion years old.
- **Customer Arrivals:** If arrivals are random (Poisson process), the time you've already waited for a customer does not change how much longer you will likely wait.

3. Normal (Gaussian) Distribution

Notation: $X \sim \mathcal{N}(\mu, \sigma^2)$

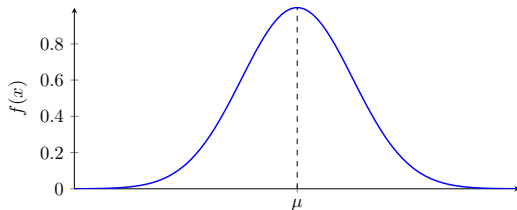
Description: The “Bell Curve”. Most common distribution in nature.

Parameters:

- μ (Mean/Location): This represents the center or peak of the “bell”
- σ^2 (Variance/Scale): This represents the “spread” of the distribution.

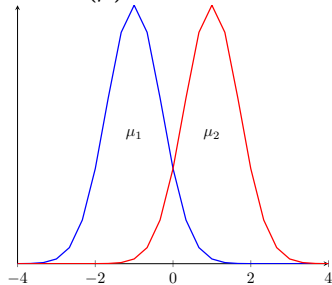
PDF:

$$f(x \mid \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

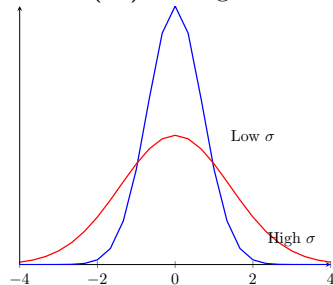


Normal Distribution: Parameters

Mean (μ): shifts the center.



Variance (σ^2): changes the width.

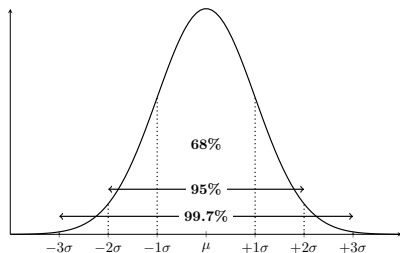


The 68-95-99.7 Rule (Empirical Rule)

For a Normal Distribution:

- **68%** of the data falls within **one** standard deviation of the mean ($\mu \pm 1\sigma$).
- **95%** of the data falls within **two** standard deviations ($\mu \pm 2\sigma$).
- **99.7%** of the data falls within **three** standard deviations ($\mu \pm 3\sigma$).

Interpretation: It is extremely rare (only 0.3% chance) for data to fall outside the 3σ range.



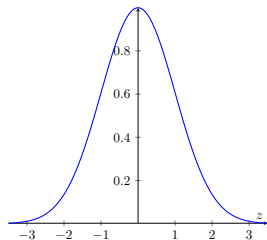
Standard Normal Distribution (Z)

If we scale any Normal variable, we get the **Standard Normal**:

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$$

Properties:

- Mean $\mu = 0$
- Std Dev $\sigma = 1$
- PDF: $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$
- Used for “Z-Scores” (how many sigmas away is a data point?).



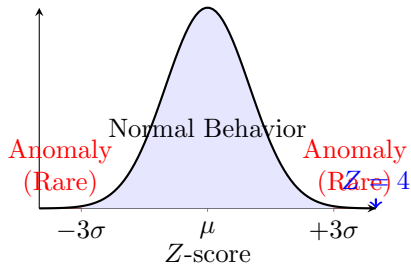
Anomaly Detection using Standard Normal Distribution

Concept:

- In a Normal distribution, values far from the mean (μ) are rare.
- We calculate the **Z-score**: $Z = \frac{x - \mu}{\sigma}$.
- **Threshold**: Typically, if $|Z| > 3$ (more than 3 standard deviations away), the data point is flagged as an **Anomaly** or **Outlier**.

Example: Network Security

- Normal traffic: $\mu = 50$ Mbps, $\sigma = 5$ Mbps.
- **Event**: Current traffic is 70 Mbps.
- $Z = \frac{70 - 50}{5} = 4$.
- Since $4 > 3$, this is an **Anomaly** (Potential DDoS attack).



Numerical Problem 1: Manufacturing

Scenario: A machine cuts rods. Length $L \sim \mathcal{N}(10\text{cm}, 0.04\text{cm}^2)$ ($\sigma = 0.2$). Rods must be between 9.6 and 10.4. What % are defective?

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Solution:

- ① Limits are 10 ± 0.4 .
- ② Since $\sigma = 0.2$, the limits are exactly $\mu \pm 2\sigma$.
- ③ We know 95% are inside $\pm 2\sigma$.
- ④ Therefore, $100\% - 95\% = 5\%$ are defective.

Numerical Problem 2: Credit Card Fraud Detection

Scenario: A bank's fraud detection system monitors the spending habits of a customer. Their transaction amounts follow a Normal Distribution with a mean $\mu = \$50$ and a standard deviation $\sigma = \$15$.

The system is programmed to flag any transaction with a Z-score greater than 3 ($|Z| > 3$) as “**Suspicious Activity**”.

Question: A new transaction of \$100 is attempted on the card. Will the system flag this transaction?

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Solution:

① Calculate the Z-score:

$$Z = \frac{x - \mu}{\sigma} = \frac{100 - 50}{15} = \frac{50}{15} \approx 3.33$$

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Solution:

- 1 Calculate the Z-score:

$$Z = \frac{x - \mu}{\sigma} = \frac{100 - 50}{15} = \frac{50}{15} \approx 3.33$$

- 2 **Compare with Threshold:** Since $3.33 > 3$, the transaction is more than 3 standard deviations above the average spending.
- 3 **Conclusion:** The system **will flag** this transaction as suspicious.

Numerical Problem 3: Inverse Problem

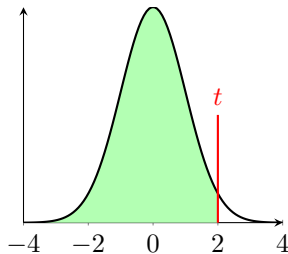
Scenario: Delivery time $T \sim \mathcal{N}(30, 16)$ ($\sigma = 4$). We want to promise a time t such that we are 97.5% sure we arrive before t .

Numerical Problem 3: Inverse Problem

Scenario: Delivery time $T \sim \mathcal{N}(30, 16)$ ($\sigma = 4$). We want to promise a time t such that we are 97.5% sure we arrive before t .

Solution:

- We need the cumulative area to be 0.975.
- 95% is middle. The tails are 2.5% each.
- Area to left of $+2\sigma$ is
50%(left half) + 47.5%(half of 95) = 97.5%.
- $t = \mu + 2\sigma = 30 + 2(4) = 38$ mins.



Chapter 5: Quiz

Quiz: Question 1

Q1: Can the value of the Probability Density Function $f(x)$ be greater than 1?

- A) No, probability can never exceed 1.
- B) Yes, because $f(x)$ is density, not probability.

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- A) No, probability can never exceed 1.
- B) Yes, because $f(x)$ is density, not probability.

Answer: B. $f(x)$ represents density. For example, if $X \sim U(0, 0.1)$, the height is $1/(0.1) = 10$. The **Area** must be 1, but height can be anything.

Quiz: Question 2

Q2: Calculate $E[X]$ if $f(x) = 3x^2$ for $0 \leq x \leq 1$.

A) 0.5 B) 0.75 C) 1.0 D) 0.25

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A) 0.5 B) 0.75 C) 1.0 D) 0.25

Answer: B) 0.75

$$E[X] = \int_0^1 x(3x^2)dx = \int_0^1 3x^3dx = \left[\frac{3x^4}{4} \right]_0^1 = \frac{3}{4}$$

Quiz: Question 3

Q3: For a Normal distribution, what is the probability of being more than 3 standard deviations away from the mean (in either direction)?

A) 5% B) 1% C) 0.3% D) 0.0%

Quiz: Question 3

Q3: For a Normal distribution, what is the probability of being more than 3 standard deviations away from the mean (in either direction)?

A) 5% B) 1% C) 0.3% D) 0.0%

Answer: C) 0.3% 99.7% is inside, so $100 - 99.7 = 0.3\%$ is outside.

Quiz: Question 4

Q4: The time to wait for a bus is Uniformly distributed between 0 and 20 minutes ($U(0, 20)$). What is the variance?

- A) 100 B) 10 C) 33.33 D) 20

Quiz: Question 4

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A) 100 B) 10 C) 33.33 D) 20

Answer: C) 33.33

$$Var(X) = \frac{(b-a)^2}{12} = \frac{(20-0)^2}{12} = \frac{400}{12} \approx 33.33$$

Quiz: Question 5

Q5: You are comparing two Normal Distributions: $X \sim \mathcal{N}(0, 1)$ and $Y \sim \mathcal{N}(5, 100)$. Which variable is more likely to produce a value close to its mean?

- A) X (Standard Normal)
- B) Y (Normal with Variance 100)
- C) Both are equal

Quiz: Question 5

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- A) X (Standard Normal)
- B) Y (Normal with Variance 100)
- C) Both are equal

Answer: A) X. Y has a variance of 100 ($\sigma = 10$), meaning it is very spread out (flat). X has variance 1 ($\sigma = 1$), meaning it is tightly clustered around the mean.

- **Continuous RVs:** Probabilities are Areas (Integrals).
- **PDF vs CDF:** Density vs Cumulative Probability.
- **Distributions:**
 - Uniform (Equal likely range).
 - Exponential (Memoryless decay).
 - Normal (Bell curve, central to statistics).
- **68-95-99.7 Rule:** Quick estimation for Normally distributed data.

Thank You!